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Assessing capacity to deploy direct air capture technology at the country level – an expert and information entropy comparative analysis

David Izikowitz¹ , Jia Li^{2,3}, Eryu Wang², Biao Zheng^{1,4} and Yuquan W Zhang^{1,4,*}

¹ China-UK Low Carbon College, Shanghai Jiao Tong University, 3 Yinlian Road, Pudong New Area, Shanghai 201306, People's Republic of China

² The Hong Kong University of Science and Technology (Guangzhou), Nansha, Guangzhou 511400, People's Republic of China

³ HKUST-Jiangmen Laboratory for Carbon Science and Technology, No. 29 Jinzhou Road, Jiangmen 529100, People's Republic of China

⁴ School of Environmental Science and Engineering, Shanghai Jiao Tong University, Shanghai 200240, People's Republic of China

* Author to whom any correspondence should be addressed.

E-mail: yqwzhang@sjtu.edu.cn

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Abstract

An ever-dwindling carbon budget, resulting in temperature rise of 1.5 °C above pre-industrial levels projected between 2030–2035, has generated a necessity to explore climate mitigation technologies such as direct air capture (DAC). DAC typically involves the use of materials and energy to capture CO₂ directly from the atmosphere. However, DAC technologies remain a long way from the necessary level of development and scale needed to move the needle on carbon removal and mitigating against climate change. This study conducts a country-level analysis using an expert elicitation and an information entropy method, with a weighted group of variables identified from existing literature as necessary to develop and deploy low-temperature, electrochemical and high-temperature DAC technologies. Here we show that: (1) adopting the expert survey variable weighting, USA, Canada, China and Australia are best positioned to deploy the various DAC technologies; (2) the information entropy approach offers a broadly similar result with traditionally developed nations being best positioned, in addition to land rich countries, to deploy DAC technologies; (3) a comparatively developed policy and financing environment, as well as low carbon energy supply would raise a country's DAC capacity; (4) developing countries such as China have significant potential to deploy DAC, owing to a well-rounded position across variables. These results produce wide-ranging policy implications for efforts to deploy climate mitigation technologies through the development of a multilateral, coordinated mitigation and carbon dioxide removal deployment strategy.

Acronyms

BECCS	Biomass Energy with Carbon Capture and Storage
BiCRS	Biomass Carbon Removal and Storage
CDR	Carbon Dioxide Removal
DAC	Direct Air Capture
DACS	Direct Air Capture with Storage
FAO	Food and Agriculture Organisation
IEA	International Energy Agency
IMF	International Monetary Fund

IPCC	Intergovernmental Panel on Climate Change
IRENA	International Renewable Energy Agency
OECD	Organisation for Economic Co-operation and Development
UN	United Nations
UNEP	United Nations Environment Program
US EIA	United States Energy Information Administration

1. Introduction

1.1. Background

The IPCC's Sixth Assessment Report (AR6) found a remaining carbon budget of 460 GtCO₂ as of January 1st, 2021, for a 50% probability of limiting warming to within 1.5 °C, and a carbon budget of 360 GtCO₂ for a higher 66% probability [1]. With annual emissions amounting to 36.4 GtCO₂ in 2021 [2], the remaining carbon budget leaves a vital decade to avert exceeding 1.5 °C of warming, which the IPCC suggests will bring increased risk to health, food security and economic well-being [3]. To put the scale of the challenge into context, in the UNEP's 2021 Emissions Gap report, the UN warned that unless emissions fall by 14% each year between 2022 and 2040, without reliance on carbon dioxide removal (CDR), the world will fail to meet the 1.5 °C ambition from the Paris Agreement [4]. Given the trajectory of emission reductions necessary to meet the 1.5 °C target, attention is increasingly turning to CDR solutions to extend the remaining carbon budget. In particular, IPCC models forecasting mitigation pathways which aim to limit warming to 1.5 °C by 2100 after a temporary temperature overshoot, have relied on assumptions of large-scale deployment of CDR measures [3].

Commonly referenced nature-based CDR solutions include the likes of afforestation, reforestation and soil carbon sequestration. Engineered CDR typically involves direct air capture (DAC), sometimes referred to as DACS to explicitly highlight the combination with durable storage. Meanwhile, hybrid CDR solutions— involving aspects of natural and engineered carbon removal—include biochar, enhanced rock weathering and biomass carbon removal and storage (BiCRS). Nature-based CDR solutions as well as BECCS, the more commonly referenced acronym for biomass energy with carbon capture and storage, have been frequently investigated. These include in many Integrated Assessment Modelling (IAM) studies [5–9], with various conclusions concerning land-use requirements and with regards to BECCS, the sustainable use of biomass. Complementary CDR techniques, including DAC, have existed for decades, however they have only recently started receiving serious attention as potential mitigation pathways to meet the goals of the Paris Agreement [10–12].

Models reviewed by the IPCC suggest cumulative carbon capture capacity required by 2100 could be between 348 and 1,218 GtCO₂ [3], and some research suggests that DAC could scale to a capacity of 30 GtCO₂/year [10, 13–16]. This brings into question the resource (energy, land, water, financial capital) demands of such a volume of DAC facilities, given that 30,000 facilities extracting 1 MtCO₂/year, or 30 million plants with a factory-manufactured, shipping container style modular configuration extracting 1,000 tCO₂/year would be required to meet this level of scale. Deployment on such a scale by the end of the century will be significant, however would by no means be unprecedented historically. This volume of deployment is comparable to the number of chemical plants in operation with 16,128 operating in just the US and Canada [17], or the production of small, standardized modular units aligned with the annual production of cars at over 70 million globally [18].

Here, an expert elicitation and information entropy analysis is used for the first time to address the following questions: (1) what characteristics would facilitate DAC to best realize its mitigation potential? And (2) assuming it's emergence as a promising CDR solution, how may DAC best be deployed across the globe? The second question may be of particular interest as it attempts to identify key countries or regions with the 'capacity' to mitigate legacy carbon emissions.

1.2. Literature review

1.2.1. DAC as a solution for decarbonisation and net zero

DAC is not a new technology, developed as early as the 1930s in cryogenic air separation plants, with the process later being applied to life support systems in space stations and submarines [19]. In contrast to carbon capture from power plants and industrial facilities, DAC enables the direct capture of CO₂ from ambient air, and was first proposed in 1999 by Klaus Lackner as a climate mitigation solution. Like other CDR solutions, DAC can address emissions from distributed sources, such as those from transport (aviation, shipping) and hard-to-abate sectors (agriculture, heavy-industry), altogether accounting for almost 50% of global emissions [20].

DAC is unique in the context of the portfolio of available CDR solutions. Some DAC technologies are able to produce a nearly pure stream of carbon dioxide, enabling multiple applications for the utilization of this CO₂; for example, in concrete, vodka, enhancing crop growth in controlled agricultural environments, carbonation of soft drinks and for those special occasions, diamonds [21, 22]. Some countries are exploring CO₂ utilization as an explicit aspect of their decarbonization strategy, seen in the launch of Saudi Arabia's 'Circular Carbon Economy National Program' in 2021 seeking to reduce, reuse, recycle and remove atmospheric CO₂ [23]. Potential CO₂ utilization applications are however limited by the type of DAC technology employed. For example, the DAC company Carbon Collect adopts a passive, humidity-swing process with an anion exchange resin typically generating a non-food grade stream of CO₂ [24]. This passive process requires a trade-off between the high energy requirements commonly required of DAC systems, at the expense of a product with lower CO₂ purity. This process can be considered more cost-effective for particular utilization applications like enhancing crop growth in controlled environments (greenhouses), but predominantly for CO₂ storage applications [12].

Further, while the mineralization of CO₂ in concrete does durably store the captured carbon dioxide, most DAC with utilization applications are likely to be, at best, carbon neutral. Therefore, in order for direct air capture to be a net-CDR solution, DAC needs to be paired with reliable long-term carbon storage mechanisms—for example, geological sequestration in disused oil wells and unmineable coal seams distributed globally, or mineralisation in rock formations [25, 26]. Notwithstanding technical considerations, significant uncertainty remains with regards to the economic viability across the various DAC technologies being explored. The DAC company Climeworks in 2019 quoted costs of US\$500–600/tCO₂, while commercial forecasts for long-term costs point to pricing of below US\$100/tCO₂ [27–29]. In spite of the uncertainties around cost and the early stage of technical development, DAC is receiving an increasing amount of attention, particularly as companies developing the technology approach commercialization.

1.2.2. Considerations for high-temperature and low-temperature DAC deployment

Taking Carbon Engineering and Climeworks as examples adopting different technical approaches to DAC, the aqueous solutions in the case of Carbon Engineering require high-temperature heat from oxy-fuel for regeneration ($T > 800$ °C), while Climeworks' amine adsorbents require 85 to 120 °C [30]. Climeworks' low-temperature process enables the use of microwave radiation as a thermal energy source, but more typically, the recovery and use of waste heat, which can reduce CO₂ capture costs by up to 40% [31]. Research suggests the thermal energy produced as waste heat is between 23%–53% of global primary energy input, with theoretical and economic recovery potential of 6%–12% and 6%–9%, respectively [32]. This wide-availability of waste heat to power low-temperature DAC processes has already been demonstrated by Climeworks in their pilot facility in Hinwil, Switzerland, co-located with a municipal waste incineration facility, and can be extended across low-temperature DAC deployment scenarios to benefit from zero cost thermal energy [33, 34]. In addition, the use of low carbon electrical energy is another attribute for DAC to be deployed commensurate with its climate mitigation intentions. A study of marginal abatement costs by Mac Dowell *et al* illustrated this such that the use of renewables for DAC with storage is more cost effective than using this energy to produce 'green' methanol to substitute gasoline [35].

Meanwhile, the high-temperatures required for current calcination-based processes have largely rendered renewable energy sources unsuitable to-date. That said, nascent, renewables powered high thermal energy systems are being developed and brought to market including resistive heating, electric arc furnaces and H₂-fired calciners. Furthermore, water availability also remains a significant constraint. An analysis of commercial plans by Viebahn *et al* found a land use requirement for high-temperature process of 0.0016 km²/MtCO₂/year - purely for the capture facility itself, not including energy generation facilities [36, 37]. In contrast, analysis of DAC using a low-temperature sorbent process found a much higher land use requirement of 0.1 km²/MtCO₂/year, again not accounting for energy generation facilities to power the process [36]. With regards to the water demands of DAC, these largely depend on the process. Stolaroff *et al* suggested a high-temperature calcination-based process would require a significant 8.18 m³/tCO₂ captured, meanwhile figures from an operational pilot facility found water loss of 4.7 m³/tCO₂ [38, 39]. In contrast, an amine-based, as well as a zeolite and metal-organic framework (MOF) physio sorbent-based process, typically produces water, with the vapor taken in through the capture of ambient air amounting to the production of approximately 1 m³/tCO₂ [36].

1.2.3. Considerations for electrochemical DAC deployment

Electro-swing DAC technologies have received heightened attention since the publication of widely-publicized research by Voskan and Hatton, leading to the formation of the DAC company Verdox to commercialize the technology. Their process involves an electrochemical cell, which utilizes a redox reaction upon the change of electric potentials for the adsorption and desorption of CO₂, simply requiring electricity to power the process [40]. Additionally, because the method only uses electrical energy, it can be switched on and off with little loss in

operating efficiency, which makes it suitable for deployment with intermittent renewable energy sources, albeit with capital cost and asset utilization implications [41]. As advantages, the electro-swing technologies from Verdox and RepAir avoid the use of heat, pressure and water, while modularity and a combination with electrochemical utilization of CO₂ is feasible. Similar to other DAC technologies, the electrochemical DAC system is also influenced by the availability of low carbon energy capacity and geological conditions. Since most electro-swing methods still need to advance beyond lab scale, financial capital and policy support are also critical for novel electrochemical DAC technologies to achieve commercialization.

1.3. Research scope

Summarizing the above literature review, there is currently a significant gap in the literature exploring the geophysical and resource variables contributing to the deployment of DAC technology, and on a wider basis, the development of a global commercial DAC industry. The research concentrated on the non-technical aspects of DAC has focused on the economics of the technology at a facilities level, along with its potential role in climate change mitigation [5, 19, 28, 30]. Few studies have explored how this technology may best be deployed in practice on a broader scale. This study attempts to overcome the constraints for developing and deploying large volumes of DAC facilities at the country level. Taking both resource and economic aspects into consideration, this work evaluates the necessary variables and addresses the gaps surrounding these variables towards the commercialization of DAC.

This study develops an evaluation system based on variables identified in existing literature. It employed the support of international experts in DAC technology, in conjunction with an information entropy analysis, to determine weights for those variables. The different DAC technologies explored, combined with utilization or storage, require unique deployment conditions. These different DAC technologies are likely to be optimal for different countries, with the evaluation carried out to deliver a comprehensive country-by-country analysis for the deployment of 6 different types of DAC solutions. These include:

- (1) Low- temperature DAC with Storage (LT-DAC and S),
- (2) Low-temperature DAC with Utilization (LT-DAC and U),
- (3) High-temperature DAC with Storage (HT-DAC and S),
- (4) High-temperature DAC with Utilization (HT-DAC and U),
- (5) Electrochemical DAC with storage (EC-DAC and S), and
- (6) Electrochemical DAC with Utilization (EC-DAC and U).

While acknowledging moisture-swing DAC approaches are being explored, including commercially, this study focused on those most widely considered. Thus, a moisture-swing DAC approach was not explored in this analysis. The outline of variables used in the expert survey and information entropy analysis for the six types of DAC included in this study are outlined in figure 1. Forthcoming sections of this paper outline the methodology used in the study—in particular, the two alternatives of weights employed in determining the indexes (section 2). Results of the analysis are then summarized with respect to each country's comparative level of development across the two methods (section 3). This paper then elaborates on the potential policy implications (section 4), and concludes with a discussion of limitations and potential future research (section 5).

2. Methodology and data

This study intends to assess, incorporating an index construction methodology, the necessary variables to deploy DAC technologies on a country-by-country basis.

Given data constraints, not every country is covered globally. It is also necessary to recognize that there is intra-country regional heterogeneity, for example with regards to renewable energy deployment or CO₂ storage infrastructure, however it is the intention of this study to provide a country-level foundation with which to build on for future analysis. This framework is established to evaluate each country's position from the perspective of their capacity to establish DAC and potentially other carbon removal solutions. The necessity to examine resource constraints for the deployment of DAC technologies, in addition to considerations around data availability prior to conducting the analysis, have been discussed in section 1. Data for each variable was collected, based on present day values from various authoritative sources, summarized in table 1. The data used in this study underwent a process of normalization to obtain appropriate scores with which to generate an index in combination with the derived weights.

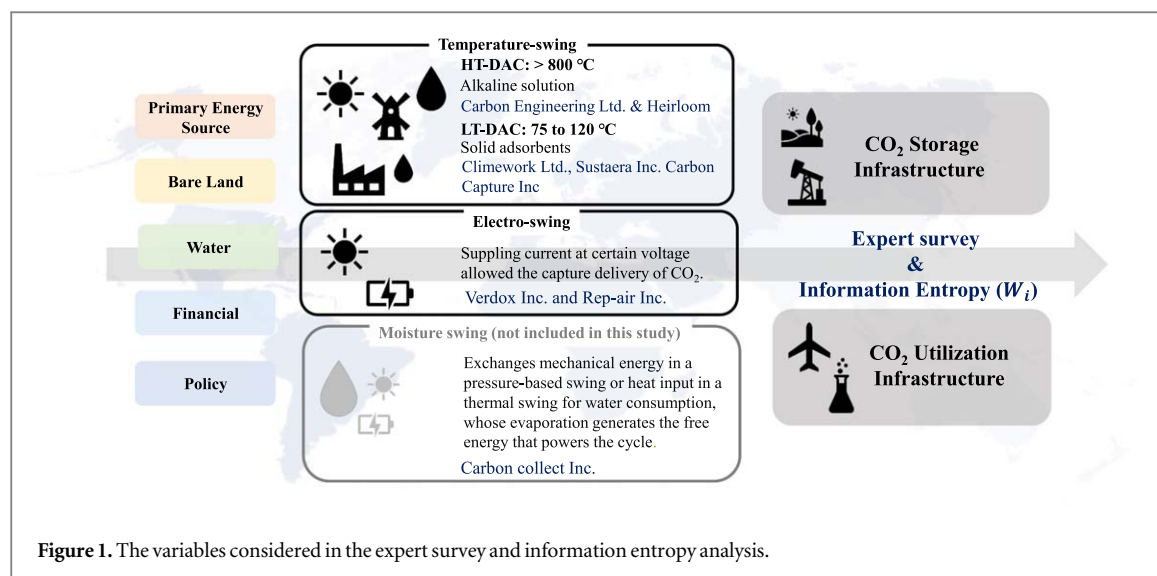


Figure 1. The variables considered in the expert survey and information entropy analysis.

Table 1. The variables for the deployment of different types of DAC technologies.

Variables	DAC Types	References
Primary Energy Source	HT-DAC, LT-DAC and EC-DAC	[30, 36, 42]
(Waste Heat, Low Carbon, Fossil Energy)	HT-DAC, LT-DAC and EC-DAC	[30, 36, 42, 43]
Land	HT-DAC	[30, 36]
Water	HT-DAC and LT-DAC	[5, 19]
Financing	HT-DAC, LT-DAC and EC-DAC	[7, 8, 42, 44]
Policy	HT-DAC, LT-DAC and EC-DAC	[44–46]
Storage Infrastructure	HT-DAC, LT-DAC and EC-DAC	[44, 47]
Utilization Infrastructure	HT-DAC, LT-DAC and EC-DAC	[44, 47]

2.1. Data review

2.1.1. Low carbon energy

The availability of low carbon energy was determined by taking a measure of the primary energy consumption from low carbon energy sources, against the total for each country. Low carbon energy sources in this context include all non-fossil sources - nuclear, hydro, wind, solar, geothermal, biomass in power generation and other renewables. This data was obtained for the 2021 annual period, with the percentage of low carbon energy taken as a proportion of total energy used as a measure of the availability of low carbon energy with which to operate DAC technologies in this study [48].

2.1.2. Policy environment

The policy environment for direct air capture is rapidly emerging and evolving. Given the significance of policy for decision-making purposes in the context of DAC, a proxy measure for the development of the policy environment for DAC at a country-level was sought and obtained courtesy of the Global CCS Institute and their CCS Policy Indicator (CCS-PI) [49]. The CCS-PI tracks the development of government policy to accelerate the implementation of carbon capture, adopting a quantitative methodology from 32 factors combining to form nine policy measures including: policy leadership, government commitment, fiscal incentives, information sharing and adoption, regulation, public finance, international collaboration, market mechanisms and institutional strengthening. This measure, the latest of which is from 2018, is by no means perfect in the constantly evolving DAC technology policy landscape. That said, it was evaluated as the best available proxy measure for a developed carbon capture policy environment for point-emission sources to extend to applications in DAC technologies, with each country's score out of 100 adopted for use in this study.

2.1.3. Land

European Space Agency satellite data collated by the Organisation for Economic Co-Operation and Development (OECD) provided a measure of each country's land composition [50]. This data, the latest of which is available from 2015, included a measure of 'bare land'. While acknowledging the location flexibility of some DAC technologies adopting a small, standardized, modular design - for example for co-location amongst

the built environment, the benefits often touted of DAC compared to biological CDR methods involve not competing for arable land, with this study seeking to evaluate the availability of said land area. As a result, the measure of bare land was used as a proxy measure for unutilised land which was available to deploy DAC technology. The percentage total for bare land area was normalized to generate a score out of 100 for use in the study.

2.1.4. Financing

Given the variety of financing mechanisms used for DAC technologies to date, it is challenging to generate a measure of the availability of financing. The deployment of DAC technologies at any form of scale is dependent on project financing, and thus a measure of infrastructure finance was sought to evaluate the availability of financing for DAC technology deployment. A recently released 'Green Infrastructure Financing Propensity Index', published by the global professional services firm PwC, was adopted due to the inclusion of a number of attributes including green financing opportunities, green growth, commitment to green objectives, the strength of the financial market and the regulatory, business and macroeconomic environment [51]. The country-level results from this index were normalized to a measure out of 100 for use as the availability of financing measure in this study.

2.1.5. CO₂ storage infrastructure

Direct air capture, in absence of safe and durable CO₂ storage, can never be carbon dioxide removal on a lifecycle basis. As such, the consideration of the development of CO₂ storage infrastructure is a vital component of this study in the context of DAC deployment with CO₂ storage. A measure was derived from the CO₂ Storage Resources Management System (SRMS) - a globally recognized evaluation methodology for CO₂ storage infrastructure development evaluated into six categories: Undiscovered Inaccessible, Prospective, Discovered Inaccessible, Contingent, Capacity, Stored [52]. While countries often possess a variety of CO₂ storage reservoirs at various stages of maturity, this study adopted a linear measure with a country given a score from 0–5 based on the most advanced level of CO₂ storage infrastructure development at the country-level, then normalized to a result out of 100 for use in this study.

2.1.6. CO₂ utilization infrastructure

The development of CO₂ utilization infrastructure is a valuable attribute given the potential pathways of CO₂ utilization in accelerating decarbonization towards net zero. While no comprehensive measure of the development of a CO₂ utilization ecosystem exists, a proxy measure was sought to indicate country-level capacity for CO₂ utilization from DAC technologies. This measure was derived from the 2022 dataset of the BP Statistical Review of World Energy, taking ethanol production data at a country-level as a proxy for the development of CO₂ utilization infrastructure capacity [53]. Ethanol production was determined as the best available proxy data source for CO₂ utilization infrastructure, given the material supply-chain and chemical processing involved in feedstock conversion to an ethanol product. This ethanol production data was normalized by taking the percentage of global production at the country-level, before a transformation via $\log(1 + X)$ to achieve a representative input of capacity.

2.1.7. Waste heat

The availability of waste heat is a critical consideration for the majority of DAC systems currently being developed. Recognizing the use of waste heat would typically involve the co-location of DAC with industrial facilities, a measure of waste heat was derived from an average of the country-level share of production of steel, cement and nuclear power [54–56]. As for the prior treatment of a country share, each measure was transformed via $\log(1 + X)$ before taking an average of the three results to derive a waste heat measure for the study.

2.1.8. Fossil energy

While low carbon energy is seen as an imperative for the majority of operations of a DAC facility, high-temperature DAC processes, today, require the use of fossil energy to achieve temperatures to sustain such thermal processes. An average of natural gas and coal production was sought from the latest BP Statistical Review of World Energy dataset to evaluate the country-level capacity to power a high-temperature DAC process, while recognizing these fuel inputs are both globally traded commodities [53]. The country-level share was normalized again taking $\log(1 + X)$ to standardize the input data before producing an average of natural gas and coal production figures to generate a measure for the study.

2.1.9. Water

Data for a measure of water availability utilized the World Resources Institute (WRI) Aqueduct 3.0 model water risk framework as a proxy for water availability on a national level. This framework combines a total of 13 water

Table 2. Breakdown of experts' sectoral background participating in the expert elicitation survey.

Professional Background of Survey Respondents	Number of Respondents
Academic researching direct air capture technology	17
Working for a startup on direct air capture technology	13
Working in industry on direct air capture technology	8
Working for a funder looking at direct air capture technology	6
Working for a non-profit looking at direct air capture technology	4
Independent expert looking at direct air capture technology	4
Other (scoping the DAC industry)	2

risk indicators—including physical risk quantity, physical risk quality, and regulatory and reputational risks—into a composite overall water score [57]. Countries are given a score out of 5 by the Aqueduct framework, which was then reversed to demonstrate lower scores as representing a greater water risk, and later normalized to scores out of 100 for the purpose of this study.

2.2. Expert survey method

An expert elicitation approach was used to determine the respective weights given to each of the variables, across the various DAC technologies and applications, to ultimately construct an index of countries based on their capacity to establish DAC technology. Variables to consider for each DAC deployment scenario were informed by a review of previous research (table 1), with a normalized score used against the weights derived from the expert elicitation to construct an index for each DAC technology and deployment scenario. This expert elicitation process took the form of a survey, distributed globally, concentrated in 'the west', to a number of participants and ecosystem actors in the DAC field from the authors' network across the industry. These experts were identified on the basis of their leading role in the commercialisation of DAC technology, or their academic credentials in the research of DAC technology. A summary of survey participants and their respective role in the DAC industry is summarized in table 2.

A weighted average was taken of the 54 responses to the survey to determine the weights given to each of the variables analyzed in the expert survey approach. Figure 2 shows the weighting factors for each of the variables for all six of the DAC technology and deployment scenarios. The respective variable weights derived from the expert survey, in combination with the variable score itself, produce a country-level index to gauge the relative capacity to deploy each of the DAC technologies.

2.3. Information entropy method

The concept of information entropy was first introduced by Claude Shannon, a mathematician and cryptographer, in his seminal paper of 1948 titled 'A Mathematical Theory of Communication' [58]. Entropy can be considered as a measure of average uncertainty, such that when all outcomes are equally likely, entropy is at its maximum.

Applied as a secondary analysis to evaluate the country-level capacity to deploy DAC technologies, this approach is advantageous given that it is able to quantify a qualitative indicator while not requiring an individual decision-maker—as necessary with an expert survey-based approach—to weigh the variables [59]. An entropy weight method has been used in multiple studies previously, particularly as a variable weighting technique [60–62], considered in this study seeking to gauge the uncertainty in each variable in their necessity for the deployment of DAC technologies.

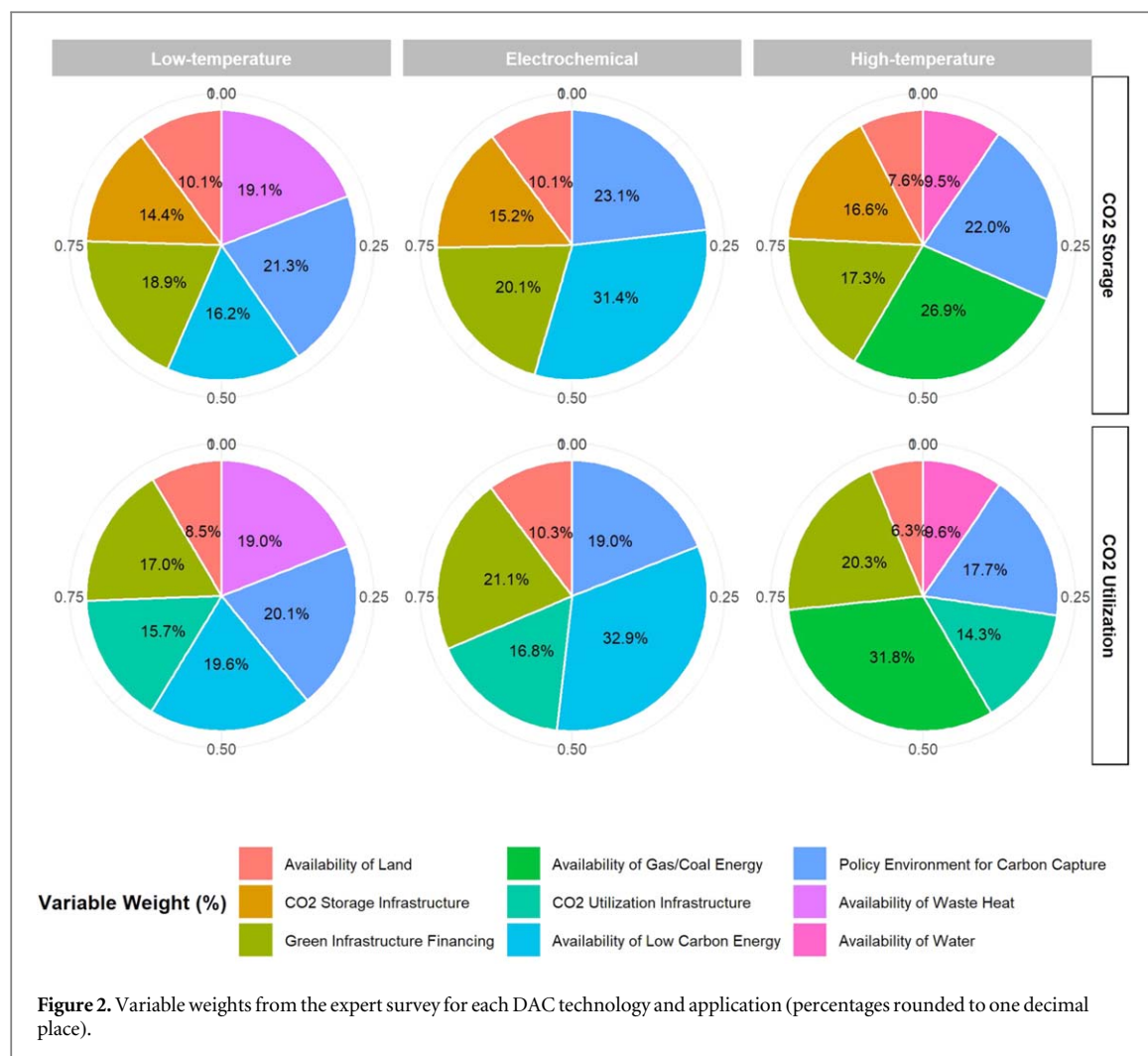
2.3.1. Calculation of indicator weights

The entropy, or uncertainty, intrinsic to each variable is calculated from the normalized variables considered in the expert survey analysis.

$$x_{new} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x is the given datapoint, x_{min} is the minimum value in the variable dataset and x_{max} is the corresponding maximum value in the variable dataset to obtain the x_{new} value.

The information entropy for each of the study variables is then calculated using an equivalent method adopted by Nie and Zhang [60]:



$$INFO_i = -k \sum_{j=1}^m f_{ij} \ln f_{ij} \quad (2)$$

where $f_{ij} = \frac{q_{ij}}{\sum_{j=1}^m q_{ij}}$ with q_{ij} representing variable i in country/region j , and $k = \frac{1}{\ln m}$ where there are n variables reflecting DAC deployment for a number of m countries/regions. The entropy value w_i can thus be determined as:

$$w_i = \frac{1 - INFO_i}{\sum_{i=1}^n (1 - INFO_i)} \quad (3)$$

Using the above formulas, entropy weights corresponding to each variable are generated and shown in table 3. Likewise, with the expert survey method, the entropy weights are used alongside the corresponding country-by-country data for each of the relevant variables to generate an index based on the information entropy in this dual method analysis.

3. Results

3.1. Expert survey

3.1.1. Low-temperature DAC with storage

The study results offer a number of key insights. Across the various DAC technologies, the broadly comparable weighting in the survey results given to the primary energy source, policy environment, financing, storage and utilization infrastructure is reflected in the results, as shown in figure 3. For example, for low-temperature DAC with storage, the USA and Canada - with a comparatively developed policy environment, financing and CO₂ storage infrastructure, translate into a high position in the index, placed #1 and #2, respectively. This dynamic is reflected in most countries evaluated as well positioned to deploy low-temperature DAC with storage, but for China. China is uniquely placed for low-temperature DAC deployment, owing to its outsized position in waste

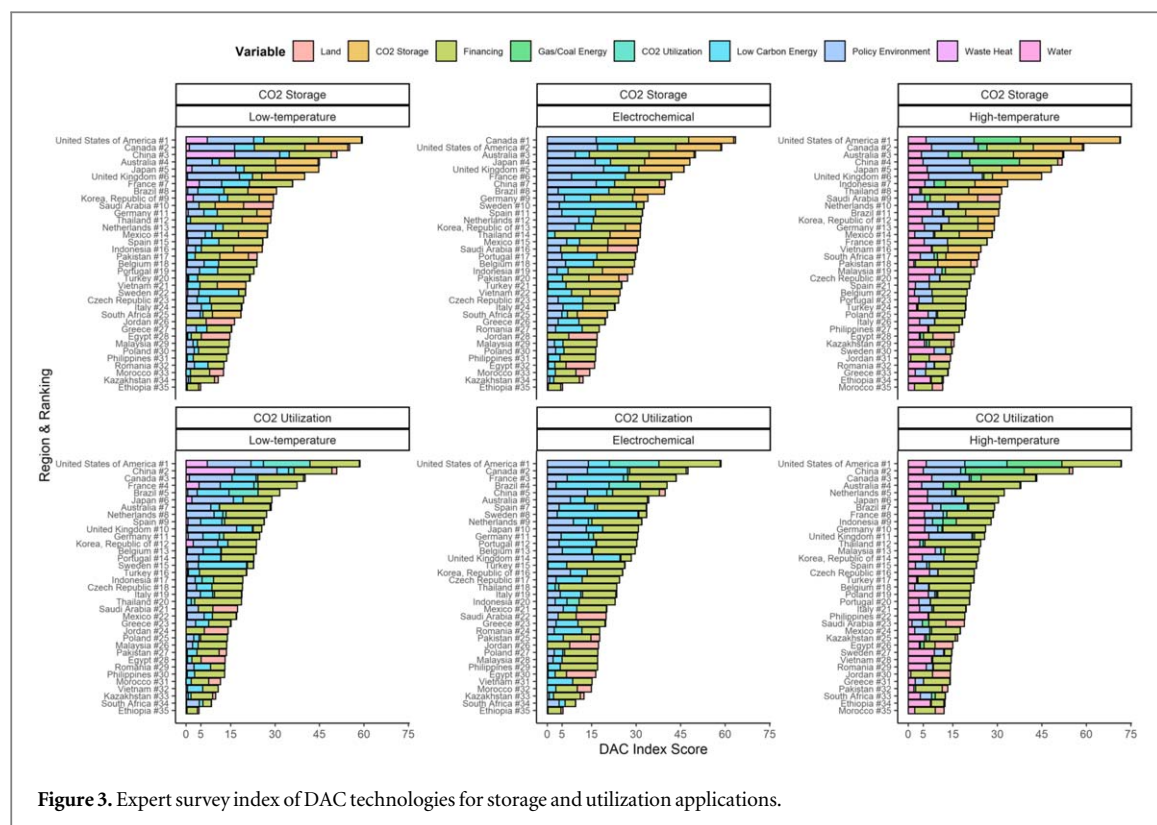


Figure 3. Expert survey index of DAC technologies for storage and utilization applications.

Table 3. Information entropy weight values for each variable across all DAC technologies and applications.

Low-temperature DAC	With Storage	With Utilization
Waste Heat	0.289059	0.250612
Low Carbon Energy	0.063427	0.054991
Land	0.317099	0.274922
Finance	0.041149	0.035676
Policy	0.102297	0.088691
CO ₂ Storage	0.186968	0.295109
Electrochemical DAC		
Low Carbon Energy	0.089216	0.073381
Land	0.446027	0.366862
Finance	0.057880	0.047607
Policy	0.143890	0.118351
CO ₂ Storage	0.262987	0.393800
High-temperature DAC		
Gas/Coal Energy	0.309639	0.268106
Water	0.036535	0.031635
Bare Land	0.320190	0.277242
Finance	0.041550	0.035977
Policy	0.103295	0.089439
CO ₂ Storage	0.188791	0.297600

heat through its global dominance in steel and cement production, alongside a significant and growing nuclear power fleet. This availability of waste heat for the co-location of DAC at industrial and nuclear power facilities in China is reflected in its position in the low-temperature DAC index, even considering the present state of its comparatively underdeveloped CO₂ storage infrastructure.

3.1.2. Low-temperature DAC with utilization

Considering low-temperature DAC with utilization, the results broadly reflect those for CO₂ storage applications. While traditionally developed economies still dominate the top of the index, Brazil's position

having leapfrogged Japan, Australia and the UK from the low-temperature with storage index, can largely be attributed to its developed CO₂ utilization infrastructure. Similar to the impact of China's outsized waste heat capacity, Brazil's dominance in ethanol production is a significant contributor to its position in the index, notwithstanding Brazil's solid positioning in low carbon energy with a hydropower dominated energy system. A noteworthy result is found towards the bottom of the index in the case of Romania and The Philippines achieving broadly the same result from diverging inputs. While Romania's place in the index is largely a result of a nearly equal contribution from its policy environment, low carbon energy and financing position, The Philippines is heavily skewed towards its position on financing to achieve broadly the same final result.

3.1.3. Electrochemical DAC with storage

The weighting for electrochemical DAC with storage is heavily oriented towards the availability of low carbon energy - accounting for almost a third of the total variable weight, which produces some noteworthy results. While the leading nations in the index are reflective of those from the low-temperature DAC results, the improved positioning of Sweden is particularly notable. The high weighting of low carbon energy combined with Sweden's hydro and nuclear power dominated energy system produces a scenario for Sweden to be well positioned to deploy electrochemical DAC, albeit with a lack of CO₂ storage infrastructure at present. While the availability of low carbon energy is a significant contributor in this index from the expert survey derived weights, it is by no means all-encompassing. This is seen in the case of Thailand, with minimal low carbon energy resources, largely compensated for by the presence of CO₂ storage infrastructure and a productive financing environment to be in a position only slightly below Sweden in the index.

3.1.4. Electrochemical DAC with utilization

A similar dynamic is found in the results of the electrochemical DAC with utilization index to that from low-temperature DAC with utilization. Besides Sweden replacing the UK in the electrochemical DAC index, all of the top 10 nations in the index are the same. This reflects the impact of the survey weights with the combined primary energy weights for low-temperature DAC with utilization (38.6%), aligned with that for electrochemical DAC with utilization (32.9%). Furthermore, while the availability of land was not broadly considered a key variable of consideration given the low weighting across DAC technologies, the highest land weighting from the survey was given for electrochemical DAC with utilization and does influence the index construction. This can be seen in the context of Saudi Arabia possessing the largest measure of bare land, compensating for its lack of low carbon energy to be positioned in the median of the electrochemical DAC with utilization index.

3.1.5. High-temperature DAC with storage

The high-temperature DAC with storage index largely follows similar patterns with the alternative DAC technologies. This is largely a result of the USA, Canada and Australia comparatively well positioned across the variables of consideration, including those specifically relevant to conduct high-temperature DAC with respect to the availability of water and fossil energy resources. The presence of developing countries like Indonesia and Thailand near the top of the index further suggests that countries with well-rounded attributes across all variables considered, as opposed to countries with extremely positive attributes on a few measures e.g. The Netherlands, represent improved capacity to deploy DAC technology especially on a localized level.

3.1.6. High-temperature DAC with utilization

The high-temperature DAC with utilization index also exhibits similar characteristics of prior DAC technologies for utilization applications. The loss of the variable for CO₂ storage infrastructure is beneficial for a number of countries which are consistently positioned higher for utilization compared to storage applications, namely China, France and The Netherlands. A noteworthy result is observed in the context of Sweden which is well positioned to deploy electrochemical DAC, however finds itself in the bottom quartile of the high-temperature DAC with utilization index, lacking in all variables but for water availability in this deployment scenario. Similarly, Saudi Arabia and its relative capacity to deploy high-temperature DAC, albeit with consideration required around water availability, does decrease significantly when considering utilization compared to storage applications based on the variable weights. This differential highlights the importance of each variable in the index construction with a change from storage to utilization applications leading to a fall in the index from position #9 to #23.

Table 4. Entropy index of low-temperature DAC with storage and utilization.

Rank	Country	Storage Index	Country	Utilization Index
#1	Saudi Arabia	47.11	United States of America	50.89
#2	United States of America	43.36	China	40.30
#3	China	42.54	Saudi Arabia	30.04
#4	Canada	36.00	Egypt	27.69
#5	Japan	32.25	Jordan	27.24
#6	Egypt	31.93	Brazil	25.12
#7	Jordan	31.42	France	17.58
#8	United Kingdom	30.37	Canada	16.19
#9	Australia	29.48	Morocco	14.52
#10	Pakistan	25.10	Japan	11.75
#11	Brazil	20.19	Indonesia	11.57
#12	Mexico	19.28	Pakistan	10.95
#13	Korea, Republic of	17.65	United Kingdom	10.66
#14	Indonesia	17.38	Germany	10.07
#15	France	17.30	Netherlands	10.03
#16	Thailand	17.03	Korea, Republic of	9.90
#17	Morocco	16.75	Spain	9.56
#18	South Africa	16.23	Australia	9.46
#19	Vietnam	16.12	Sweden	8.08
#20	Germany	14.51	Belgium	7.53
#21	Netherlands	9.21	Italy	7.16
#22	Spain	9.04	Portugal	6.69
#23	Sweden	8.63	Thailand	6.11
#24	Belgium	8.12	Mexico	5.91
#25	Portugal	7.40	Turkey	5.71
#26	Italy	6.94	Kazakhstan	5.47
#27	Turkey	6.58	Czech Republic	5.46
#28	Kazakhstan	6.30	Greece	4.80
#29	Czech Republic	6.30	Poland	4.67
#30	Greece	5.54	Romania	3.88
#31	Romania	4.48	Malaysia	3.60
#32	Poland	4.20	South Africa	3.27
#33	Malaysia	4.16	Vietnam	3.17
#34	Philippines	3.54	Philippines	3.07
#35	Ethiopia	3.11	Ethiopia	2.70

Notes: The backgrounds of developed countries and the associated scores are shaded.

3.2. Information entropy

3.2.1. Low-temperature DAC with storage and utilization

Adopting an information entropy approach yields somewhat different results to that observed through the expert survey approach.

Principally, an entropy approach produces contrasting variable weights for low-temperature DAC (table 4) with regards to the availability of land. A land weighting of 31.7% for low-temperature DAC with storage and 29.5% for low-temperature DAC with utilization produce some diverging results when compared to the expert survey approach. The significant land weighting is reflected with the positioning of Saudi Arabia, Egypt and Jordan all above the likes of the UK and Australia; owing to the minimal weighting given to financing (4.1%) and the policy environment (10.2%) in the low-temperature DAC with storage index.

This dynamic with conflicting weights to the expert survey method is largely continued in the low-temperature DAC with utilization index. CO₂ utilization infrastructure (29.5%), the availability of land (27.5%) and waste heat (25.1%) form the vast majority of the index construction. This is reflected in the countries best positioned to deploy low-temperature DAC with utilization in the study, with the USA and Brazil possessing the most developed infrastructure for CO₂ utilization. Meanwhile, Saudi Arabia, Egypt and Jordan host the highest availability of land, and China is best positioned on the basis of waste heat availability. These attributes contribute to their respective high-performing position in their capacity to deploy low-temperature DAC with utilization on the basis of an information entropy analysis.

3.2.2. Electrochemical DAC with storage and utilization

The contrasting variable weights for the entropy approach compared to the expert survey method continue in the index construction for electrochemical DAC (table 5).

The significant weighting given to the availability of land (44.6%) in the electrochemical DAC with storage measure, alongside CO₂ utilization infrastructure (39.4%) in the electrochemical DAC with utilization index, is similar to the dynamic found with low-temperature DAC. This produces broadly aligned index results with those from the low-temperature DAC entropy weighted indexes. Land-rich countries (Saudi Arabia, Egypt, Jordan) and those with developed CO₂ utilization infrastructure (USA, Brazil) are well positioned to deploy electrochemical DAC according to the entropy-derived weights.

3.2.3. High-temperature DAC with storage and utilization

A theme of contrasting weights from the expert survey approach continues in the entropy method for high-temperature DAC with storage and utilization (table 6).

The variable weights largely reflect those derived in the low-temperature DAC entropy index. The primary energy source accounts for 31.0% of the index weight for high-temperature DAC with storage, and 26.8% for high-temperature DAC with utilization. This produced comparable results with the alternative DAC technologies considering the weight also given to the availability of land with the USA, Saudi Arabia, Canada, China and Australia all well positioned on the basis of the entropy weights for high-temperature DAC with storage.

While the weights are significantly different to those derived from the expert survey, the final result is similar with seven of the top ten in the high-temperature DAC with storage index shared by both the expert survey and entropy derived weights. This is replicated in a high-temperature with utilization deployment context with six of the top ten positions shared by both weight-derivation methods, with the difference a result of the outsized weight allocated to the availability of land.

3.3. Economic and regional analysis

An in-depth review and graphical representation of the results, presented through a measure of economic development by developed and developing countries, is offered in figures 4 and 5. Overall, the results indicate that developed countries possess a notably larger capacity to deploy DAC technologies than developing ones. This is observed when evaluating figure 4, conducting a regression of the index scores across DAC technologies and applications plotted against GDP per capita. The regression analysis presents a clear positive correlation - especially when adopting an expert survey approach, between index scores and GDP per capita, particularly pronounced in the context of electrochemical DAC. This indicates that, as is often observed with the diffusion of technologies across history, that developed countries are at present better positioned to deploy DAC technologies considering the parameters evaluated in this study.

Similar observations can be drawn from the box plots in figure 5, albeit with some caveats. The range for developed countries across DAC technologies with storage as the end CO₂ state is significantly broader for both the expert survey and entropy approaches than that observed across the board for developing countries and for utilization applications. This indicates that while the general trend observed is that developed countries are better positioned to deploy DAC technologies, it is by no means a rule. Country-specific considerations are thus needed beyond the question of 'developed' or 'developing' country.

Differences can also be observed between the expert survey and entropy approaches. It is clear from figure 5 that entropy weights tend to reduce the index score across all DAC technologies compared to expert survey index

Table 5. Entropy index of electrochemical DAC with storage and utilization.

Rank	Country	Storage Index	Country	Utilization Index
#1	Saudi Arabia	65.76	United States of America	55.30
#2	Canada	48.22	Saudi Arabia	39.67
#3	United States of America	45.65	Egypt	36.59
#4	Egypt	44.49	Jordan	36.33
#5	Jordan	44.17	Brazil	32.50
#6	United Kingdom	41.60	China	25.03
#7	Australia	41.31	Canada	19.62
#8	Japan	41.11	Morocco	19.30
#9	Pakistan	34.43	France	15.71
#10	Brazil	27.16	Indonesia	15.05
#11	Mexico	26.26	Pakistan	13.90
#12	China	24.89	United Kingdom	13.31
#13	Indonesia	23.97	Netherlands	13.23
#14	Thailand	23.62	Australia	12.49
#15	Morocco	23.46	Germany	12.23
#16	South Africa	22.38	Japan	12.18
#17	Vietnam	21.70	Spain	11.59
#18	Korea, Republic of	19.62	Sweden	9.81
#19	Germany	18.95	Belgium	9.28
#20	France	14.92	Italy	9.17
#21	Netherlands	12.76	Korea, Republic of	8.93
#22	Spain	11.29	Portugal	8.87
#23	Sweden	10.96	Thailand	7.88
#24	Belgium	10.48	Mexico	7.18
#25	Portugal	10.34	Kazakhstan	7.18
#26	Italy	9.30	Turkey	6.72
#27	Kazakhstan	8.73	Czech Republic	6.69
#28	Turkey	8.17	Greece	6.35
#29	Czech Republic	8.13	Poland	6.02
#30	Greece	7.73	Romania	4.91
#31	Romania	5.97	Malaysia	4.62
#32	Poland	5.65	South Africa	3.99
#33	Malaysia	5.62	Philippines	3.94
#34	Philippines	4.79	Ethiopia	3.54
#35	Ethiopia	4.31	Vietnam	3.43

Notes: The backgrounds of developed countries and the associated scores are shaded.

results, especially considering developed countries for CO₂ utilization applications. This can likely be explained by the outsized weighting given to the availability of land with developed countries generally poor positioned with respect to the availability of land, which contributes to the dynamic where developed countries appear relatively worse positioned to deploy DAC technologies when considering the entropy approach.

Table 6. Entropy index of high-temperature DAC with storage and utilization.

Rank	Country	Storage Index	Country	Utilization Index
#1	United States of America	51.72	United States of America	58.19
#2	Saudi Arabia	49.81	China	36.12
#3	Canada	38.35	Saudi Arabia	32.23
#4	China	37.73	Egypt	29.54
#5	Australia	35.77	Jordan	27.74
#6	Egypt	34.12	Brazil	24.61
#7	Jordan	32.03	Canada	18.05
#8	United Kingdom	31.35	Indonesia	16.33
#9	Japan	30.89	Morocco	14.80
#10	Pakistan	24.98	Australia	14.73
#11	Indonesia	22.93	Netherlands	11.57
#12	Brazil	19.56	United Kingdom	11.34
#13	Mexico	19.36	Pakistan	10.73
#14	Thailand	18.51	France	10.45
#15	South Africa	18.44	Japan	10.40
#16	Vietnam	17.24	Germany	9.79
#17	Morocco	17.09	Kazakhstan	7.87
#18	Korea, Republic of	14.77	Spain	7.55
#19	Germany	14.23	Korea, Republic of	7.34
#20	Netherlands	10.98	Thailand	7.30
#21	Kazakhstan	9.09	Malaysia	7.13
#22	France	9.07	Italy	7.06
#23	Malaysia	8.24	Poland	6.68
#24	Czech Republic	6.94	Belgium	6.05
#25	Italy	6.83	Czech Republic	6.01
#26	Spain	6.71	Mexico	5.87
#27	Poland	6.51	Portugal	5.79
#28	Belgium	6.41	Sweden	5.70
#29	Portugal	6.37	Ethiopia	5.14
#30	Ethiopia	5.93	Turkey	5.08
#31	Sweden	5.89	South Africa	5.07
#32	Turkey	5.87	Philippines	4.46
#33	Philippines	5.16	Greece	4.33
#34	Greece	5.01	Romania	4.28
#35	Romania	4.94	Vietnam	4.03

Notes: The backgrounds of developed countries and the associated scores are shaded.

Similar observations can be drawn from the regional clustering analysis illustrated in figure 6. The arrows in the figure representing movement in index scores between the expert survey method and entropy approach, further evidence the effect of the availability of land weight with Middle East and North Africa countries with an improved positioning on the basis of the entropy approach, while all other regions largely fell. In addition,

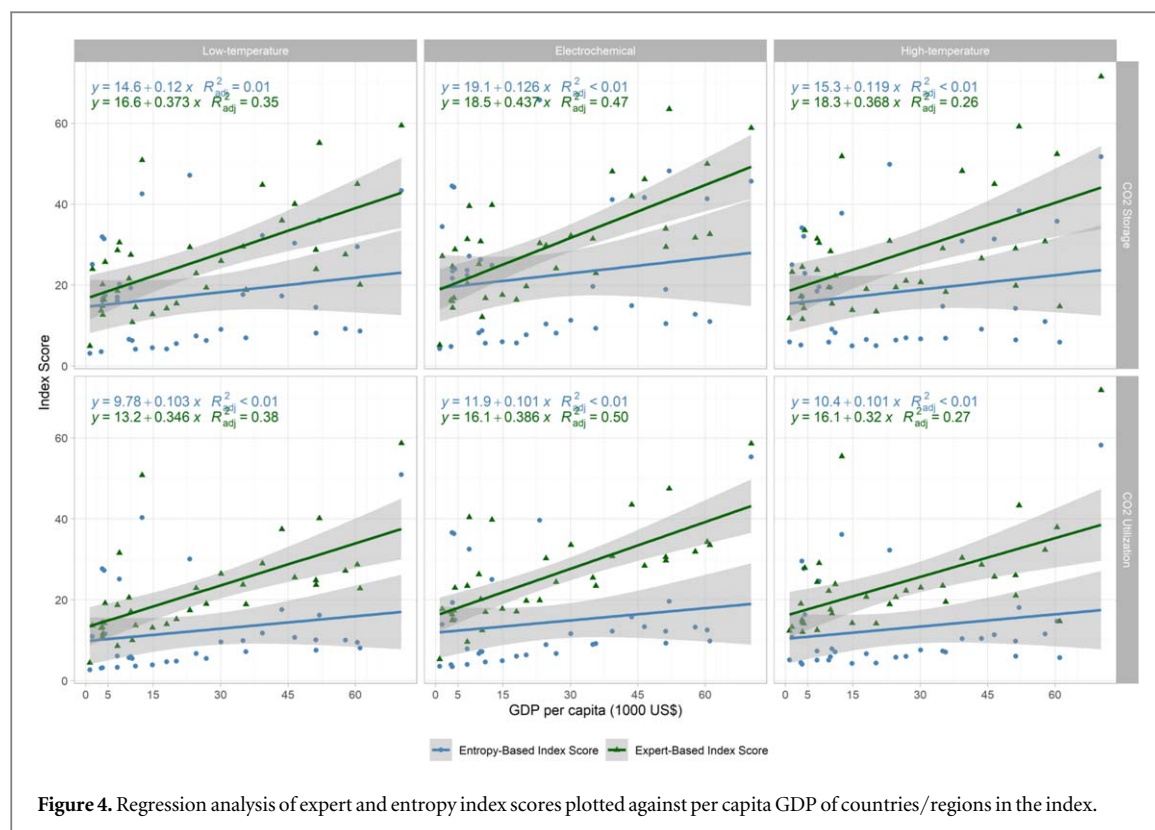


Figure 4. Regression analysis of expert and entropy index scores plotted against per capita GDP of countries/regions in the index.

observing the arrow gradient further indicates the disparity between expert and entropy index results, with the most extreme changes seen by Canada, Mexico and China across utilization applications; largely a product of the disparity in weights given to financing and the policy environment from the expert survey method compared with their minimal share for the entropy method.

4. Discussion and future research

This analysis builds on established literature which has identified the scientific necessity to develop and deploy carbon removal solutions like DAC. To limit temperature increases to the Paris Agreement's 2 °C goal, annual emissions in 2030 should be 15 GtCO₂ lower than the current unconditional targets set out in each country's Nationally Determined Contributions (NDCs), or a much more ambitious reduction of 32 GtCO₂ to meet the more stringent 1.5 °C goal [4]. This raises the question of ambition, on a government level, to achieve what the science suggests is necessary. Assessments are increasingly including carbon capture in mitigation pathways, alongside the fact that the level of ambition in NDCs must increase at least threefold for 2 °C and fivefold for the 1.5 °C goal [3].

Notes: The shaded section illustrates the area falling into the 0.95 confidence interval.

Notes: Diamond dots indicate group means, cross dots denote outliers, boxes represent the range between the 25th to 75th percentile, lines within the box indicate the median, and the ends of lines outside of boxes indicate 1.5 IQR.

4.1. Large demand for low carbon energy

While DAC technologies have a land and occasionally a water footprint, there are fewer natural resource constraints that may limit the deployment of DAC compared to the likes of BECCS and other biological CDR pathways [30]. However, with regards to energy constraints, the operation of a DAC facility as a carbon removal solution is heavily reliant on the use of zero-carbon energy, and in the case of heat-powered DAC technologies, thermal energy to power a DAC unit. The deployment of DAC at scale may thus significantly impact global energy provision placing additional demand on already constrained renewable energy capacity—standing at only 38.3% of total electricity generating capacity in 2021 [63], which will need to grow at a largely unconstrained rate to both decarbonize energy generation and power a climate consequential DAC industry.

Various theoretical and practical solutions to this power conundrum exist, for example the utilization of curtailed renewable energy generated in periods of excess supply, as well as innovation in sorbent technologies

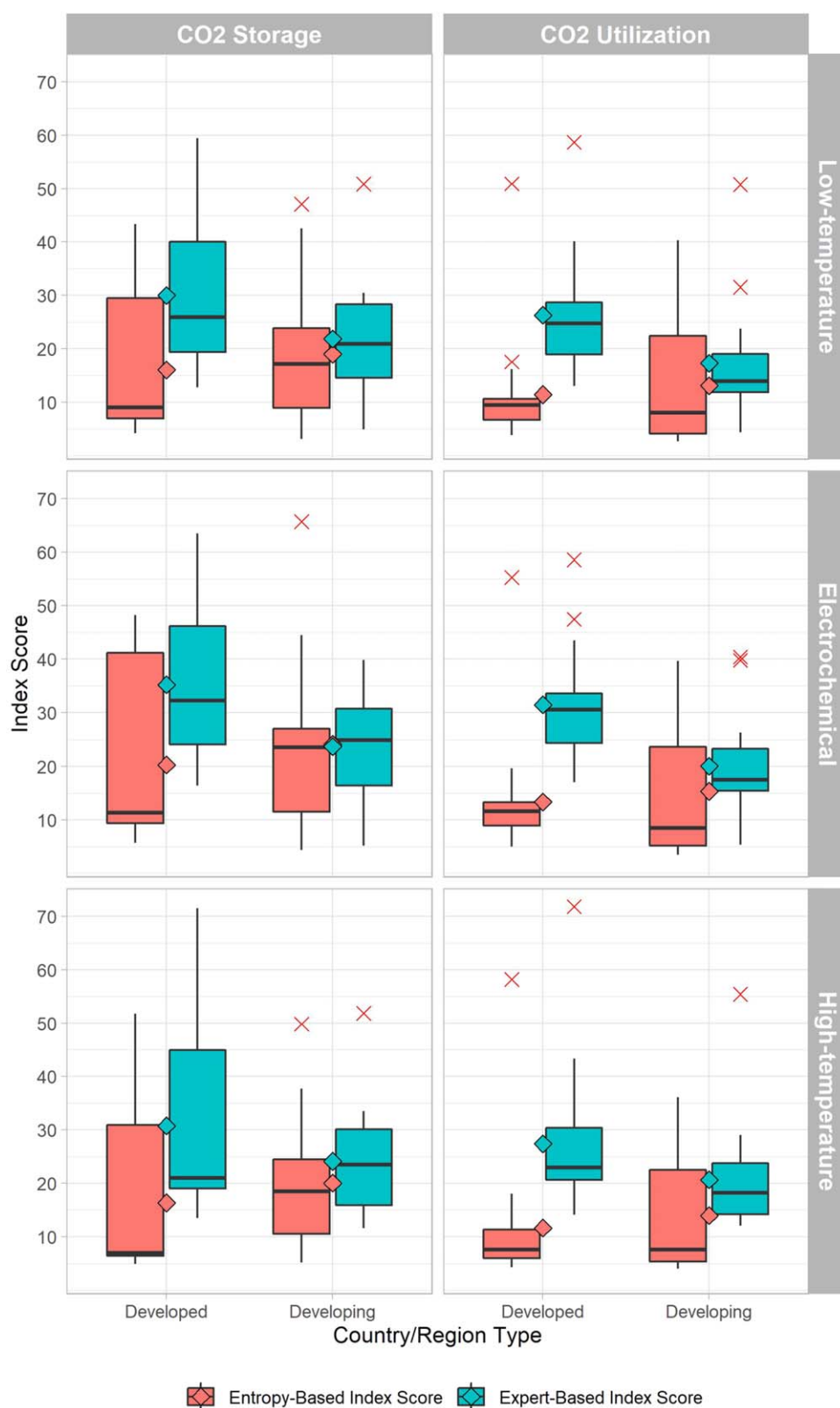


Figure 5. Box-plots of expert and entropy index scores for all DAC technologies between developed and developing countries/regions.

optimized to operate in different conditions [64]. This would raise additional questions as to the asset utilization of a DAC facility reliant on the use of excess electricity, as has occasionally been discussed particularly in the context of electrochemical DAC deployment, however this could be mitigated through the deployment of dedicated energy storage infrastructure. On the other hand, a high-temperature, hydroxide solution absorption and calcination-based (liquid-solvent) process, like that currently adopted by the Canadian DAC company

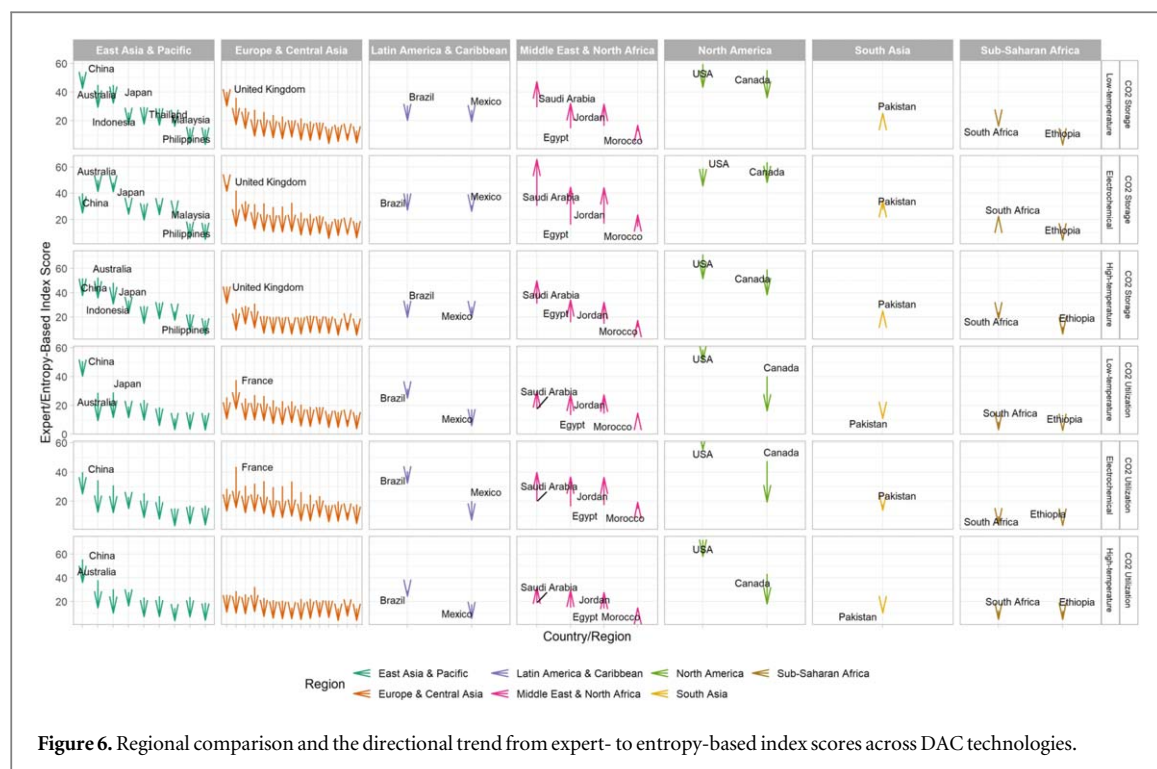


Figure 6. Regional comparison and the directional trend from expert- to entropy-based index scores across DAC technologies.

Carbon Engineering, requires continuous temperatures of up to 900 °C which will be generated through the combustion of fossil energy, namely natural gas [39]. This brings rise to prospective questions around fugitive emissions upstream and the lifecycle emissions associated with a fossil energy powered DAC process. Future innovation in low emission heat generation could address the high-temperature requirement of calcination-based DAC processes.

4.2. Financial capital from the private sector

It is important to identify the public-private sector dynamic in the future development and deployment of technologies like DAC and the broader suite of climate change mitigation and CDR solutions. This study derived its financing measure for green infrastructure predominantly taking into consideration private sector factors including green financing opportunities, green growth and the macroeconomic, regulatory and business environment. It is often a combination of public capital (US Department of Energy, Canada Strategic Innovation Fund, EU Commission Grants) and that of private sector capital (corporate innovation funds, venture capital, private infrastructure funds) which provide the financing to bring technologies like DAC to market. Such private sector sources of financial capital are largely unconstrained by national borders, potentially plugging gaps where nation states have insufficient financial capital to invest in the pioneering R and D necessary to commercialize DAC. This can be observed with private sources of capital across the DAC commercialization spectrum. From the early stage venture capital investor Lowercarbon Capital and their US\$350 million fund dedicated to CDR, to Frontier Climate's US\$925 million CDR purchasing facility and the US\$4.5 billion Global Infrastructure Fund IV by Blackrock, all with an openly global funding remit [65–67]. This DAC-eligible private sector financing is complemented by emerging public funding mechanisms, largely taking the form of policy measures, accounted for in this study through the Global CCS Institute's Policy Indicator. For example, the recent revision of the 45Q tax credit in the US Inflation Reduction Act provides additional incentives for eligible DAC projects conducting CO₂ storage or particular CO₂ utilization applications which meet specific requirements [68]. The financing environment for DAC deployment, spanning both private and public sources of capital, is nascent but rapidly emerging with regular new funding announcements contributing to broad-based attention, likely translating into capital commitments as DAC technologies achieve commercial progress and begin deployment [69].

4.3. Differentiated capacity in carbon removal

Given the lack of international progress on global emission reduction since the Paris Agreement of 2015, an alternative approach may be appropriate to break the impasse and unlock inter-governmental progress and ambition. Climate change mitigation debates frequently focus on the question of responsibility, rather than one of capacity. In the meantime, there is no escaping the reality that current unconditional NDCs put the world on a path to a temperature increase of 2.6 °C, far exceeding the goals of the Paris Agreement [4].

Table 7. Top 10 cumulative CO₂ emitters and corresponding average of expert-based index score.

Country/region	Cumulative CO ₂ emissions (Billion Tonnes)	Expert index score
United States of America	399.38	63.14
China	200.14	48.07
Russia	100.59	Insufficient Data
Germany	90.57	28.83
United Kingdom	77.07	35.16
Japan	62.3	38.51
India	48.56	Insufficient Data
France	37.77	35.67
Canada	31.91	51.44
Ukraine	25.8	Insufficient Data

Orienting towards a question of *capacity*, rather than responsibility, yields some noteworthy insights as can be seen in table 7 showing the top 10 cumulative CO₂ emitters alongside their respective average scores across the six DAC technologies and applications in the expert-based study. While the USA is responsible for 25% of global cumulative CO₂ emissions to date, our study results suggest it is best placed across an average of the six DAC technologies and applications considered in this study. Meanwhile, other major emitters like China and Canada are also well positioned, with France similarly placed to the UK, on the basis of capacity from our study [70]. The recent extension of the government led international climate innovation agency Mission Innovation with a new ‘mission’ to advance carbon dioxide removal is a signal of this capacity framing. The initiative, co-led by the USA, Canada and Saudi Arabia, with additional involvement from Japan, Australia, Norway, India, the UK and European Commission, reflects the results and ambitions of this study to inform which countries possess a comparative advantage in the deployment of DAC technologies [71].

4.4. Research limitations

While this study seeks to compare countries on their current position to deploy DAC technologies through variables evaluated in the literature review, these variables are by no means exhaustive, nor all encompassing. This is largely a result of data limitations. For example, considerations around the development of sorbent material supply-chains - which ought to be explored in future iterations where openly available data exists. Furthermore, the breadth of data availability was largely limited to high- and middle-income countries, which also ought to be addressed in future research. Notwithstanding these limitations surrounding data depth and breadth, this study aims to offer a valuable foundation from which to build on in evaluating the current country-level capacity to deploy a selection of DAC technologies currently being commercialized.

4.5. Future research

Further to the aforementioned data constraints, additional analysis could be conducted by the research community. Future studies could aim to build on the approach and country-level ranking conclusions outlined in this study as new data sources become publicly available. While this high-level approach is a constructive foundation for the deployment of various DAC technologies, the DAC research community would benefit from a further granular analysis at the localized, intra-country level to explore the necessary attributes for site selection in the context of real-world DAC deployment. This would be especially valuable given the wide variety of DAC technologies being proposed and explored, for which local ambient temperature, humidity and other local variables will affect the operating performance of DAC technologies. Improved insight and awareness of such considerations would bring significant value for the DAC technology research community. Additionally, a consideration of the trend rate in the development of the variables considered, for example low-carbon energy or CO₂ storage infrastructure, is worth exploration given the extensive lead time experienced in the development and deployment of DAC technologies.

Furthermore, the availability and maturity of CO₂ storage sites is a variable worth further, in-depth investigation, in addition to a broader evaluation of CO₂ utilization infrastructure. This would be particularly valuable from the perspective of the development of a ‘circular carbon economy’ with regards to potential CO₂ utilization. For example, through the development of the US’ US\$3.5 billion ‘DAC Hubs’ and the UK’s £1 billion fund for ‘CCUS Clusters’ [72, 73]. The exclusion or inclusion of different variables, e.g. sorbent production supply-chains, or the increased prominence of certain variables (financing versus CO₂ storage infrastructure) is heavily dependent on data availability and thus has a bearing on the variables producing the index result. For

example, in absence of data availability constraints, future research could adopt a similar method to conduct a multidimensional study into DAC sorbent production. These would explore the country-level characteristics necessary to secure a vital component of a commercial DAC system and further enhance the literature available into the capacity of various geographies to develop and deploy DAC technologies.

5. Conclusions

This study identified various attributes highly relevant for the development and deployment of DAC technologies, and created various indexes of countries based on their suitability from these attributes. We find that: (1) a comparatively strong position with regards to low carbon energy, financing and the policy environment generally increases a country's DAC deployment capacity; (2) traditional developed nations like the USA, Canada, and Australia are currently best placed to advance DAC technologies; (3) developing countries like China also have a high capacity to deploy DAC technologies, again due to a developed financing, policy environment and other positive properties within the index.

Building on existing literature which principally focused on the technical and policy aspects of DAC technology, this study has broad applications and potential geopolitical implications into the discussion around climate change mitigation and the future development pathways of DAC, other carbon dioxide removal and mitigation options. In particular, given the dwindling carbon budget combined with insufficient global commitments, carbon removal solutions like DAC with CO₂ storage will increasingly become a necessity, with the results of this study acting as an ad hoc foundation for an international dialogue focused on respective national capacity to develop climate change mitigation solutions like direct air capture.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

ORCID iDs

David Izikowitz  <https://orcid.org/0000-0002-4666-1256>

Yuquan W Zhang  <https://orcid.org/0000-0001-7636-9451>

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